

Mapping MATLAB Steps to Textbook Laplace Transform Solution

Explanation of Declared Variables and Code Structure

Declared Symbolic Variables

```
syms X x(t) s
```

- $x(t)$ Represents the unknown function $x(t)$ in the differential equation. MATLAB treats it as a symbolic function of the variable t , allowing differentiation and Laplace transforms.
- X Represents the Laplace transform of $x(t)$, i.e.

$$X = X(s) = \mathcal{L}\{x(t)\}.$$

MATLAB uses X as the algebraic placeholder after transforming the ODE into the s -domain.

- s The Laplace-domain variable. After transformation, all derivatives of $x(t)$ become algebraic expressions in s , such as

$$\mathcal{L}\{x'(t)\} = sX(s) - x(0).$$

Code Block Structure and Meaning

Block 1: Define the ODE

```
eqn = diff(x, t, 2) + 3*diff(x, t, 1) + 2*x == t;
```

This corresponds to the textbook equation

$$x''(t) + 3x'(t) + 2x(t) = t.$$

Block 2: Apply the Laplace Transform

```
L_eqn = laplace(eqn, t, s);
```

This matches the textbook step

$$\mathcal{L}\{x''\} + 3\mathcal{L}\{x'\} + 2\mathcal{L}\{x\} = \mathcal{L}\{t\}.$$

Block 3: Display Laplace Transform Rules

```
L{ x(t) }    = X(s)
L{ x'(t) }   = s*X(s) - x(0)
L{ x''(t) }  = s^2*X(s) - s*x(0) - x'(0)
L{ t }       = 1/s^2
```

These are the same formulas used in the textbook.

Block 4: Substitute $\mathcal{L}\{x(t)\} = X$

```
L_eqn = subs(L_eqn, laplace(x(t), t, s), X);
```

This converts the transformed ODE into an algebraic equation in X .

Block 5: Substitute Initial Conditions

```
L_eqn = subs(L_eqn, [x(0), subs(diff(x(t), t), t, 0)], [0, -2]);
```

This corresponds to the textbook substitution

$$x(0) = 0, \quad x'(0) = -2.$$

Block 6: Solve for $X(s)$

```
X_sol = solve(L_eqn, X);
```

This matches the textbook step where the equation is solved for $X(s)$.

Block 7: Inverse Laplace Transform

```
x_t = ilaplace(X_sol, s, t);
x_t = simplify(x_t);
```

This corresponds to the textbook final solution

$$x(t) = -\frac{3}{4} + \frac{1}{2}t - e^{-t} + \frac{7}{4}e^{-2t}.$$

Verification of Block Order

The order of the blocks is correct:

1. Define the ODE.
2. Apply Laplace transform.
3. Display transform rules.
4. Substitute $X(s)$.
5. Substitute initial conditions.
6. Solve for $X(s)$.
7. Apply inverse Laplace transform.

This matches the exact order of the textbook solution.